Canonical README: NIHSS Post-Processing (2025)

This document is the **canonical README** for the NIHSS Post-Processing project.

It supersedes earlier versions (`post_processing_README.pdf`, `table_format_README.txt`) and unifies the pipeline description with the final table formatting specifications.

1. Overview

We are trying to classify patients' stroke severity according to the National Institutes of Health Stroke Scale (NIHSS) rubric, based on neurological exam text from within their electronic health records (EHRs). For each patient, we have exactly one EHR, and for each EHR there is one neurological exam, in addition to other, extraneous text. The pipeline has two parts: the automatic neurological exam extraction module, which takes unprocessed EHRs and aims to extract the neurological exam text; and the exam scoring module, which takes neurological exam text and assigns an itemized NIHSS score to the patient. The automatic exam extraction module is evaluated according to four token-level metrics–Jaccard score, F1 score, precision, and recall–and two semantic metrics–a Bio-BERT embedding similarity score and a ROUGE-L score. The scoring module evaluation is run in 4 folds in order to get standard deviations of evaluation metrics across folds.

This repository implements the post-processing pipeline for evaluating the exam scoring module's outputs. It produces a polished LaTeX longtable comparing model performance against both Gold and Silver standards.

- Models: LLaMA, GPT
- Prompts: Zero-Shot (ZS DA), Zero-Shot Chain-of-Thought (ZS CoT), Few-Shot (FS)
- Extraction Methods: Manual, Automatic, Full Note
- Subgroups:
  - Institution: BMC, MGB, MIMIC
  - Timing: pre-2015, post-2015
  - Adversarial: Reintroduced Abbreviations

Important: GPT Full Note runs are adversarial only. They must always be labeled as `Full Note Reintroduced Abbrev.` in the final table.

2. Directory Structure

```
project_root/
  evals_final_llama/       # Prediction CSVs for LLaMA
  evals_final_gpt/         # Prediction CSVs for GPT

  postproc_inputs/         # Static LaTeX components
    latex_preamble.tex
    latex_closing.tex
    latex_superpanel_fragment.tex.j2

  postproc_intermediate/
    mapfile.csv
    llama_tidyLong.csv
    gpt_tidyLong.csv
    matrices/
```

```
    llama/gold/{aggregate,per_fold}
    llama/silver/{aggregate,per_fold}
    gpt/gold/{aggregate,per_fold}
    gpt/silver/{aggregate,per_fold}

code/
  convert_gptOutputs.ipynb
  build_mapfile.ipynb
  build_tidyLong.ipynb
  build_matrices.ipynb
  build_finalTable.ipynb

postproc_outputs/        # Final LaTeX/PDF tables
```

3. GPT Conversion

`convert_gptOutputs.ipynb` parses raw GPT predictions into standardized CSVs.

- Predictions are parsed from the `gpt_NIHSS` column as text.
- Subscores are extracted and recomputed into totals (max = 42).
- Parsing truncates at the first "Total NIH Stroke Scale Score" to avoid duplication.
- Filenames follow:

  `{extract}_{Expanded|NoExpandedAbbrev}_250806fix_{prompt}_AggrPrompt.csv`

- Special case: Full Note NoExpandedAbbrev -¿ Note$_{p}rompt_{A}ggrPrompt.csv$

4. Tidy Long Files

Each tidyLong file contains one row per patient per prediction.
Columns include:

- Patient ID
- Predicted score (integer)
- Gold NIHSS total, Silver NIHSS total
- Fold index
- Year bin (pre2015/post2015)
- Dataset flag (0=BMC, 1=MGB, 2=MIMIC)
- Adversarial indicator (T/F)
- Model, prompt, extraction, reference
- Absolute error, squared error

Special rule: For GPT Full Note, all rows are adversarial.

5. Matrix Files

Aggregated Matrices (`aggregate/`):

`{metric}_{subgrpType}_{subgrpVal}_pretty.csv`

Per-Fold Matrices (per$_{f}old/$) :

`{metric}_{subgrpType}_{subgrpVal}_fold{n}.csv`

Metrics: qwk, pearson, icc3, mae, rmse, deltaT, w Subgroup Types: overall, inst, time, adv Subgroup Values: all, BMC/MGB/MIMIC, pre2015/post2015, abbrReintro

6. Final Manuscript Table Specification

Super-Columns (2)

- Comparison to Gold Standard
- Comparison to Silver Standard

Super-Panels (4)

- LLaMA Zero-Shot
- GPT Zero-Shot
- LLaMA Zero-Shot CoT
- GPT Zero-Shot CoT

Panels (Extraction Methods)

- Manual
- Automatic
- Full Note (GPT shown only as "Full Note Reintroduced Abbrev.")

Subpanels (Subgroups)

- Institution: BMC, MGB, MIMIC
- Timing: pre-2015, post-2015
- Adversarial: Reintroduced Abbreviations

Metrics per Cell

- Quadratically Weighted Kappa (QWK)
- Pearson correlation (r)
- Intraclass Correlation Coefficient (ICC3)
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Paired t-test (delta, t, significance stars)
- Wilcoxon signed-rank test (W, significance stars)

7. Statistical Notes

- ICC_3 (two-way mixed effects, absolute agreement, single measure): Computed across folds between predicted and reference NIHSS totals. Values range 0–1; higher values indicate greater reliability.
- Paired t-test: delta = mean(pred - ref). Stars: * $p < .05$, ** $p < .01$, *** $p < .001$.
- Wilcoxon signed-rank: Nonparametric test of symmetry. Stars use the same thresholds.

8. Formatting Conventions

- Super-panels are stacked vertically and can continue across pages.
- Visual divisions:
  - Black background + white text for panel banners
  - Dark gray background + white text for metric headers
  - Light gray separators for extraction methods
  - Lighter gray separators for subgroup headers
- Blank space:
  - 4pt after subgroup header
  - 8pt after subgroup block

9. Build Order

1. convert_gptOutputs.ipynb
2. build_mapfile.ipynb
3. build_tidyLong.ipynb
4. build_matrices.ipynb
5. build_finalTable.ipynb

10. Current Status (Sept 2025)

- GPT parsing corrected (gpt_NIHSS, duplication fix).
- Matrices consistent (model, reference, key order).
- GPT Full Note = adversarial only; labeled correctly.
- Final table structure matches formatting specification.
- ICC_3 metric integrated (two-way mixed, absolute agreement) and reflected in both large and small tables.